

On Decision Resolution in Process Control: Finite Constraint Propagation in Observed Processes

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Abstract

A decision system reads a monitor and chooses an action. If the monitor has finitely many outputs, the decision system inherits that finiteness: it can implement at most as many distinct decisions as the monitor can distinguish states. We develop the consequences of this observation into a quantitative theory.

We first replace the customary existence claim—“no finite monitor returns a point, so some positive resolution floor exists”—with an arithmetic identity. For any finite monitor with $|S|$ outputs partitioning a range of diameter D , the *mean* cell diameter is exactly $D/|S|$, and consequently at least one cell has diameter $\geq D/|S|$. This is a one-line count rather than a contradiction argument, and unlike the existential form it yields a number computable from an instrument’s specification sheet. We then prove that the existential form as usually stated is false: a finite monitor may have cells of diameter zero, so the infimum of cell diameters is not in general positive, and any theorem resting on that infimum is unsound. We give the corrected measure-theoretic statement, which bounds the cell one actually occupies rather than the smallest cell in the partition.

The central result is the *decision resolution bound*. Let J be an objective, A^* the pointwise-optimal decision rule, and A any rule that factors through a monitor $\mathcal{M}$. We prove that the worst-case optimality gap of A is at least half the *in-cell compromise modulus* $\Gamma_J(\mathcal{M})$, a quantity we construct after showing that the two obvious candidates fail. The modulus of continuity of J fails because a rule may place

its single action at a cell’s centre, so the worst state lies only half a width away; using the full width overstates the bound by 2^p for a p -th order objective. A swap-based divergence also fails, because a rule is not confined to the two states’ own optimal actions but may play a compromise optimal for neither, which can be four times better. The surviving object measures the least summed loss any single action must incur over a pair of states, and it correctly assigns zero to pairs that agree on their optimal action—something no distance-based modulus does. The modulus must be aggregated *within cells*, not globally: we exhibit a decision problem in which the global form asserts a gap of $\frac{1}{2}$ against a true gap of 0, because a wide cell need not contain a divergent pair. The bound is attained, hence tight, and it saturates: beyond a cardinality set by the problem’s geometry additional monitor states buy exactly nothing, and only cell placement remains. Read in bits, one additional bit of monitor buys a factor 2^p in achievable decision loss.

Three consequences follow. First, the Nyquist–Shannon sampling bound is the special case of the decision bound in which the objective is reconstruction error; we show that the widespread practice of writing a single inequality chaining a state-space cell width to a temporal sampling rate is dimensionally invalid, and we give the two separate bounds—one temporal, one spatial—that it conflates. Second, we correct the folklore claim that a subsystem cannot both monitor and control: as usually stated the claim is false, since a regulator that estimates nothing is unbiased by construction,

and we show the bias is properly a causal gap that afflicts *estimators of quantities the control policy changes*, is bounded below, and does not decay with sample size. Third, we correct the claim that robust agreement requires three independent sources: three suffices against crash faults *and against a naive arbitrary fault*, but four are required against an *adaptive* one, so a triplicate measurement protects against a dead or obviously broken instrument and not against a miscalibrated one whose drift mimics a genuine reading.

Every theorem is proved from stated hypotheses. A numerical protocol of fifteen checks accompanies the paper, of which three are deliberate falsification attempts against claims we assert to be false and one is a methodological control separating genuine violations from discretisation error. We report that the numerical suite refuted an earlier global form of the main theorem and forced the in-cell statement, rather than presenting the corrected theorem as though it had been the original.

Keywords: decision resolution; quantised control; rate-distortion; finite observer; resolution floor; modulus of continuity; causal bias; Byzantine agreement; measurement uncertainty.

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1 Introduction

1.1 The question

Consider any system that observes and then acts. A cluster scheduler reads queue depths and assigns tasks to processors. A bioreactor controller reads dissolved oxygen and adjusts an impeller. A screening campaign reads assay plates and selects the next library. In each case the same architecture appears: a *monitor* maps the world into a finite set of outputs, and a *decision rule* maps those outputs into actions. The monitor is finite. This is not a contingent fact about present technology; it follows from the monitor being a physical device

with bounded memory, bounded bandwidth, and bounded time. An analogue-to-digital converter with 16 bits has exactly 65,536 output states. A plate reader reports optical density to a fixed number of digits. A scheduler polls a counter of fixed width. Whatever the mechanism, the output set is finite while the quantity observed ranges over a continuum.

The question this paper answers is:

What does the monitor’s finiteness cost the decision rule?

Not: what does it cost the *estimate*. That question is answered by quantisation theory (Gray and Neuhoﬀ, 1998; Zador, 1982) and rate-distortion theory (Cover and Thomas, 2006; Shannon, 1948), which bound reconstruction error under a bit budget. Not: does a coarse monitor still permit stabilisation. That question is answered by quantised control (Brockett and Liberzon, 2000; Delchamps, 1990) and the data-rate theorems (Nair and Evans, 2004; Tatikonda and Mitter, 2004; Wong and Brockett, 1999), which give the minimum rate for stability. Our question is different and, as far as we are aware, unanswered in this form: given an arbitrary objective J that the decision rule is trying to maximise, how much achievable objective value is forfeited because the rule can only see $|S|$ distinct states?

This is the quantity a designer actually possesses a budget for. A designer does not typically choose between stability and instability; they choose between a cheaper instrument and a more expensive one, and want to know what the difference buys.

1.2 Why the existing answers do not suffice

The three literatures above each answer a neighbouring question, and it is worth being precise about the gap, because the gap is the paper’s contribution.

Quantisation theory bounds $\mathbb{E}\|q - \hat{q}\|^2$, the error in reconstructing the observed quantity. But a decision rule does not reconstruct; it acts. Two states that reconstruct badly may map to the same optimal action, in which case the reconstruction error is irrelevant; two states that reconstruct well may straddle a decision

boundary, in which case a small reconstruction error is catastrophic. The relation between reconstruction error and decision loss is mediated by the objective’s geometry, which quantisation theory does not model.

Rate-distortion theory bounds the mutual information required to achieve a given distortion. It is the right shape of answer, and our main theorem is deliberately stated to resemble it. But the classical theory takes distortion to be a fixed, typically difference-based, function of source and reconstruction. Decision loss is not of that form: it is $J(A^*(q), q) - J(u, q)$, which depends on the objective and the action set, and which vanishes on entire regions where a single action is optimal.

Quantised control establishes what coarse feedback costs in stability. Stability is a binary property; our question is quantitative and concerns the objective value at the operating point, not whether the operating point is reached.

1.3 What is proved

We prove five results.

1. **The mean-diameter identity** (Theorem 3.1). For a finite monitor with $|S|$ outputs partitioning an interval of diameter D , the mean cell diameter is exactly $D/|S|$; consequently some cell has diameter at least $D/|S|$. The proof is a summation. This replaces the existential “there is a floor $\beta > 0$ ” with the computable quantity $\beta = D/|S|$.
2. **The infimum form is false** (Theorem 3.7). A finite monitor may partition a continuum into cells one of which is a single point. Hence $\inf_i \text{diam}(C_i)$ may be zero even though $|S|$ is finite. Any argument that defines the resolution floor as the minimum cell diameter and asserts its positivity is therefore unsound, and we exhibit the counterexample. Theorem 3.10 gives the correct replacement: a bound on the cell one occupies, holding almost surely under any absolutely continuous law.
3. **The decision resolution bound** (Theorem 5.1), the central result. For any decision rule factoring through a monitor with

$|S|$ outputs,

$$\sup_q [J(A^*(q), q) - J(A_{\mathcal{M}}(q), q)] \geq \frac{1}{2} \Delta_J(\beta)$$

where Δ_J is the *decision divergence*, a modulus we define that measures how much two states disagree about which action they want. Theorem 5.7 shows the bound is attained, hence tight. Theorem 5.8 shows it saturates: beyond a cardinality set by the problem’s geometry, additional monitor states buy exactly zero.

4. **Nyquist as a special case, and a dimensional correction** (Section 7). Taking J to be negative squared reconstruction error over a bandlimited source recovers the sampling theorem. We then show that the composite inequality $\beta \leq w_{\text{cell}} \leq 1/(2B)$, which appears in the applied literature, is dimensionally invalid—its left member is a length in state space and its right member is a time—and we give the two well-formed bounds it conflates.
5. **Estimation versus regulation** (Section 8) and **redundancy** (Section 9). The monitor–control separation claim is false as usually stated and true when restricted to estimators of policy-affected quantities, where the bias is a causal gap that does not decay with sample size (Theorem 8.4). The “coherence requires three” claim is the crash-fault case; arbitrary faults require four (Theorem 9.4).

1.4 What is not claimed

We prove theorems about a formal object: a finite monitor feeding a decision rule under a stated objective. We claim no experimental validation on physical hardware. The numerical protocol of Section 10 checks the theorems against constructed instances and against two deliberate attempts at falsification; it does not constitute deployment evidence. Where a result requires an assumption we cannot discharge in general—notably the availability of the objective’s modulus—we mark it and confine its use.

We also do not treat stochastic objectives. Every result is stated for a deterministic J and a deterministic monitor. Section 11 records

what changes under noise, and why the deterministic bound remains a lower bound in expectation but ceases to be tight.

1.5 Organisation

Section 2 fixes the formal objects. Section 3 proves the floor results and the falsity of the infimum form. Section 4 constructs the compromise modulus, in terms of which the main theorem is stated, by first refuting the two natural candidates. Section 5 proves the decision resolution bound, shows it cannot be stated globally, and establishes its tightness and saturation. Section 6 reads the bound as a rate–distortion statement and derives the bit-accounting corollaries. Section 7 recovers Nyquist and corrects the dimensional error. Section 8 treats estimation versus regulation. Section 9 treats redundancy. Section 10 gives the validation protocol and results. Section 11 states limitations. Section 12 concludes.

2 Setup

We fix notation and the standing assumptions. Everything in this section is definitional; no claim is made until Section 3.

2.1 States, monitors, cells

Definition 2.1 (Observable range). The *observable range* is a set $\mathcal{Q}$ equipped with a metric ρ . Unless stated otherwise $\mathcal{Q} \subseteq \mathbb{R}^d$ is compact with $\text{diam}(\mathcal{Q}) = D < \infty$, where $\text{diam}(A) = \sup_{x,y \in A} \rho(x,y)$. In the one-dimensional case we take $\mathcal{Q} = [0, D]$.

Compactness is not essential to the floor results but simplifies the statements; $D < \infty$ is essential, and we flag where it is used.

Definition 2.2 (Finite monitor). A *finite monitor* is a map $\mathcal{M}: \mathcal{Q} \rightarrow S$ where S is a finite set, $|S| = N < \infty$. The *cells* of $\mathcal{M}$ are the non-empty preimages

$$C_s = \mathcal{M}^{-1}(s) = \{q \in \mathcal{Q} : \mathcal{M}(q) = s\}, \quad s \in S.$$

We write $\mathcal{C}(\mathcal{M}) = \{C_s : s \in S, C_s \neq \emptyset\}$ and, for $q \in \mathcal{Q}$, $\text{cell}(q)$ for the unique cell containing q .

Remark 2.3 (Why the cells partition). The cells are pairwise disjoint (a function has one value at each point) and cover $\mathcal{Q}$ (every point has a value). So $\mathcal{C}(\mathcal{M})$ is a partition of $\mathcal{Q}$ into at most N non-empty parts. This is the only structural fact about monitors we use, and it is a consequence of $\mathcal{M}$ being a function into a finite set—not an additional hypothesis. Every subsequent result about floors is a result about finite partitions of a bounded set.

Remark 2.4 (Monitors need not be measurable or continuous). We deliberately assume nothing about the geometry of the cells. They need not be intervals, connected, convex, or measurable. Real quantisers produce interval cells, but hysteretic, adaptive, or multiplexed monitors need not, and the floor results hold regardless. Where measurability is needed—only in Theorem 3.10, which integrates against a probability law—we assume it explicitly.

2.2 Actions, objectives, decision rules

Definition 2.5 (Decision problem). A *decision problem* is a triple $(\mathcal{Q}, \mathcal{U}, J)$ where $\mathcal{Q}$ is an observable range, $\mathcal{U}$ is a set of *actions*, and $J: \mathcal{U} \times \mathcal{Q} \rightarrow \mathbb{R}$ is an *objective*, bounded above, with $J(u, q)$ the value obtained by taking action u when the true state is q .

Definition 2.6 (Optimal rule and value function). The *value function* is

$$V(q) = \sup_{u \in \mathcal{U}} J(u, q),$$

and a rule $A^*: \mathcal{Q} \rightarrow \mathcal{U}$ is *pointwise optimal* if $J(A^*(q), q) = V(q)$ for all q . We assume throughout that the supremum is attained, so A^* exists; this holds whenever $\mathcal{U}$ is compact and $J(\cdot, q)$ is upper semi-continuous, and in particular whenever $\mathcal{U}$ is finite.

Definition 2.7 (Monitor-measurable rule). A decision rule $A: \mathcal{Q} \rightarrow \mathcal{U}$ *factors through* the monitor $\mathcal{M}$, written $A \in \mathcal{A}(\mathcal{M})$, if there is $a: S \rightarrow \mathcal{U}$ with $A = a \circ \mathcal{M}$. Equivalently, A is constant on every cell of $\mathcal{M}$.

This is the formal content of “the decision rule sees only what the monitor reports”. It is the single structural constraint the whole paper turns on.

Definition 2.8 (Optimality gap). For a rule A , the *pointwise gap* is

$$\text{Gap}_A(q) = V(q) - J(A(q), q) \geq 0,$$

and the *worst-case gap* is $\text{Gap}_A = \sup_{q \in \mathcal{Q}} \text{Gap}_A(q)$. The *achievable gap* at cardinality N is

$$\text{Gap}^*(N) = \inf_{\mathcal{M}: |S|=N} \inf_{A \in \mathcal{A}(\mathcal{M})} \text{Gap}_A,$$

the best worst-case gap obtainable by *any* monitor of cardinality N together with the best rule for it.

Remark 2.9 (The infimum over monitors is the point). $\text{Gap}^*(N)$ optimises over the placement of the cells as well as over the rule. A lower bound on $\text{Gap}^*(N)$ is therefore a statement that *no* monitor of that cardinality does better, however cleverly its cells are arranged. This is what distinguishes a resolution bound from an analysis of a particular quantiser, and it is why the bound of Section 5 is architectural rather than incidental.

2.3 Standing assumptions

Assumption 2.10 (Finiteness). The monitor’s output set is finite: $|S| = N < \infty$.

Assumption 2.11 (Bounded range). $\text{diam}(\mathcal{Q}) = D < \infty$.

Assumption 2.12 (Attained optimum). $V(q) = \max_{u \in \mathcal{U}} J(u, q)$ is attained for every $q \in \mathcal{Q}$.

We will state explicitly, at each result, which of these it uses. No result uses more than these three plus hypotheses stated in its own statement.

3 The Resolution Floor

This section establishes what finiteness alone implies about a monitor’s cells. The results are elementary. We give them in full because the elementary form is exactly what is needed, and because the form commonly asserted in the applied literature is, as we show, unsound.

3.1 The mean-diameter identity

Theorem 3.1 (Mean cell diameter). *Let $\mathcal{M}$ be a finite monitor on $\mathcal{Q} = [0, D]$ whose cells are intervals, with $|\mathcal{C}(\mathcal{M})| = n \leq N$ non-empty cells. Then*

$$\frac{1}{n} \sum_{C \in \mathcal{C}(\mathcal{M})} \text{diam}(C) \geq \frac{D}{n} \geq \frac{D}{N}, \quad (1)$$

with equality in the first inequality if and only if the cells tile $[0, D]$ up to endpoints. Consequently there exists a cell C with

$$\text{diam}(C) \geq \frac{D}{N} =: \bar{\beta}. \quad (2)$$

Proof. The cells are disjoint intervals covering $[0, D]$. For a disjoint family of intervals covering an interval of length D , the sum of the lengths is at least D , since Lebesgue measure is countably additive and $\lambda(C) = \text{diam}(C)$ for an interval. Hence $\sum_C \text{diam}(C) \geq \lambda([0, D]) = D$, giving the first inequality after dividing by n ; equality holds exactly when the cover is exact, i.e. the intervals tile. The second inequality is $n \leq N$. Finally, a finite family whose mean is at least D/N has a member at least D/N , which is (2). $\square$

Remark 3.2 (What this replaces). The customary argument runs: a finite monitor cannot separate all points of an uncountable range, so some cell is non-degenerate, so a positive floor exists. That establishes existence but yields no number, and—as Theorem 3.7 shows—the version of it that defines the floor as the smallest cell is false. Theorem 3.1 instead *computes*. It requires no contradiction, no appeal to cardinality, and no topology beyond additivity of length.

Example 3.3 (A number from a datasheet). A 16-bit analogue-to-digital converter spanning 0–5 V has $N = 2^{16} = 65,536$ output codes and $D = 5$ V, so

$$\bar{\beta} = \frac{5 \text{ V}}{65,536} \approx 76.3 \mu\text{V}.$$

For a uniform quantiser every cell has exactly this width, so the bound is attained everywhere. A decision that branches on a voltage difference smaller than $76.3 \mu\text{V}$ is branching on a distinction the converter did not make.

Example 3.4 (A number from a screening rig). A plate reader reporting optical density on $[0, 4]$ to three decimal places has $N = 4001$ distinguishable outputs and $\bar{\beta} \approx 10^{-3}$ OD units. Any ranking of variants whose OD differences are below 10^{-3} is a ranking of quantiser codes, not of variants.

3.2 Higher dimensions

Theorem 3.5 (Mean diameter in $\mathbb{R}^d$). *Let $\mathcal{Q} \subset \mathbb{R}^d$ be compact with Lebesgue measure $\lambda(\mathcal{Q}) > 0$, and let $\mathcal{M}$ be a finite monitor with at most N measurable cells. Then some cell C satisfies*

$$\lambda(C) \geq \frac{\lambda(\mathcal{Q})}{N}, \quad (3)$$

and consequently

$$\text{diam}(C) \geq \left(\frac{\lambda(\mathcal{Q})}{N \omega_d} \right)^{1/d} \cdot 2, \quad (4)$$

where $\omega_d = \pi^{d/2}/\Gamma(\frac{d}{2} + 1)$ is the volume of the unit ball in $\mathbb{R}^d$.

Proof. The cells partition $\mathcal{Q}$ into at most N measurable parts, so $\sum_C \lambda(C) = \lambda(\mathcal{Q})$ and some part has measure at least the mean, giving (3). For (4), the isodiametric inequality (Folland, 1999) states that among sets of a given diameter the ball has greatest volume, so a set C with $\text{diam}(C) = \delta$ satisfies $\lambda(C) \leq \omega_d(\delta/2)^d$. Combining with (3) and solving for δ gives the claim. $\square$

Remark 3.6 (The dimensional penalty). Theorem 3.5 exposes a fact that the one-dimensional statement hides: the diameter floor shrinks only as $N^{-1/d}$. Halving the linear resolution of a d -dimensional monitor costs a factor 2^d in output states. A monitor observing five channels at 8 bits each has $N = 2^{40}$ states but a linear resolution corresponding to only 2^8 per axis. In decision terms, developed in Section 5, it is the linear resolution that matters, and it is the linear resolution that is expensive.

3.3 The infimum form is false

We now show that the version of the floor most often asserted—that the *minimum* cell diameter is positive—does not follow from finiteness, and is in fact false.

Proposition 3.7 (Degenerate cells are compatible with finiteness). *There exists a finite monitor $\mathcal{M}$ on $\mathcal{Q} = [-1, 1]$ with $|S| = 3$ such that $\min_{C \in \mathcal{C}(\mathcal{M})} \text{diam}(C) = 0$. Hence finiteness of S does not imply $\inf_C \text{diam}(C) > 0$.*

Proof. Define

$$\mathcal{M}(q) = \begin{cases} -1, & q < 0, \\ 0, & q = 0, \\ +1, & q > 0. \end{cases}$$

Then $|S| = 3$, and the cells are $[-1, 0)$, $\{0\}$, and $(0, 1]$, of diameters 1, 0, and 1. The minimum is 0. $\square$

Remark 3.8 (Why this matters). The sign-with-zero quantiser of Theorem 3.7 is not pathological; it is an exact-zero detector, a standard component. Any argument of the form “let $\beta = \min_C \text{diam}(C)$; by finiteness $\beta > 0$ ” is therefore invalid, and any theorem downstream of such a definition inherits the defect. The error is subtle because pigeonhole does give a positive bound—but on the *maximum* cell, as in Theorem 3.1, not the minimum. Passing from “some cell is large” to “every cell is large” is where the argument fails, and the two statements have opposite quantifier structure.

Remark 3.9 (What survives). Two repairs are available and we use both. The first is Theorem 3.1: bound the *mean*, which is what (1) does, and accept an existence claim about some cell. The second is to bound the diameter of *the cell one occupies*, which is the operationally relevant quantity and is the subject of Theorem 3.10. Neither repair requires the false claim.

3.4 The cell one occupies

For decisions, what matters is not the smallest cell in the partition but the cell containing the state at hand. The following gives the correct probabilistic statement.

Theorem 3.10 (Occupied-cell bound). *Let $\mathcal{Q} = [0, D]$, let $\mathcal{M}$ be a finite monitor with at most N measurable interval cells, and let μ be a Borel probability measure on $\mathcal{Q}$ that is absolutely continuous with respect to Lebesgue*

measure with density f satisfying $f \leq M$ point-wise. Then for $q \sim \mu$,

$$\mathbb{P}_\mu \left[\text{diam}(\text{cell}(q)) \geq \frac{D}{2N} \right] \geq 1 - \frac{MD}{2}. \quad (5)$$

In particular, if μ is uniform on $[0, D]$ (so $M = 1/D$),

$$\mathbb{P} \left[\text{diam}(\text{cell}(q)) \geq \frac{D}{2N} \right] \geq \frac{1}{2}. \quad (6)$$

Proof. Let $\mathcal{C}_{\text{small}} = \{C : \text{diam}(C) < D/(2N)\}$ be the small cells. Since there are at most N cells in total,

$$\lambda \left(\bigcup_{C \in \mathcal{C}_{\text{small}}} C \right) = \sum_{C \in \mathcal{C}_{\text{small}}} \text{diam}(C) < N \cdot \frac{D}{2N} = \frac{D}{2}.$$

Hence, by absolute continuity and the density bound,

$$\mathbb{P}_\mu[\text{cell}(q) \in \mathcal{C}_{\text{small}}] = \int_{\bigcup \mathcal{C}_{\text{small}}} f \, d\lambda \leq M \lambda \left(\bigcup \mathcal{C}_{\text{small}} \right) < \frac{M}{2}.$$

Taking complements gives (5). For μ uniform, $f \equiv 1/D$ so $M = 1/D$ and $MD/2 = 1/2$, giving (6). $\square$

Remark 3.11 (Reading the constant). The factor $\frac{1}{2}$ in $D/(2N)$ is a design choice trading the threshold against the probability; replacing it by $\theta \in (0, 1)$ gives $\mathbb{P}[\text{diam}(\text{cell}(q)) \geq \theta D/N] \geq 1 - \theta MD$. The content is that a monitor cannot have most of its probability mass in degenerate cells: the exact-zero detector of Theorem 3.7 has a degenerate cell, but that cell is hit with probability zero under any absolutely continuous law. *This is the precise sense in which the floor is real:* not that every cell is wide, but that the cells one actually lands in are wide with high probability.

3.5 Two floors, distinguished

The literature applies the word “floor” to two different quantities with different warrants and different units. We separate them, because theorems depending on one should not be discarded when a reader rejects the other.

Definition 3.12 (Instrumental floor). The *instrumental floor* of a monitor is $\beta_{\text{inst}} = D/N$, computed from the monitor’s output cardinality and the range it spans. It has the units

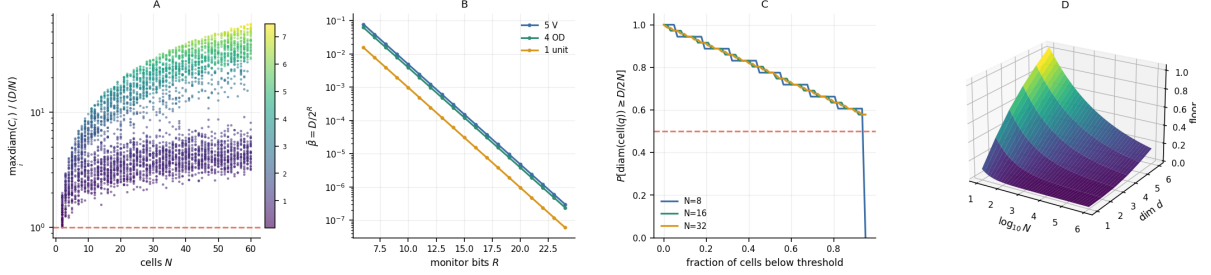


Figure 1: The Resolution Floor. (A) Largest cell diameter normalised by the floor, $\max_i \text{diam}(C_i)/(D/N)$, on a logarithmic y -axis against cell count $N \in \{2, \dots, 60\}$, for 4000 random partitions of an interval; colour encodes the coefficient of variation of the cell widths, from uniform (dark) to strongly clustered (light). Every point lies at or above unity (coral dashed), as Theorem 3.1 requires; the measured minimum of the mean ratio over 50,000 partitions is 1.000000000000. Uniform partitions sit exactly on the line—the identity is attained, not merely respected—while the excess grows monotonically with unevenness, exceeding $40\times$ for the most clustered cuts. The bound constrains the largest cell, never the smallest, which is the quantifier structure Theorem 3.7 turns on. (B) Instrumental floor $\bar{\beta} = D/2^R$ on a logarithmic y -axis against monitor bit depth $R \in \{6, \dots, 24\}$, for three operating ranges: 5 V (an analogue-to-digital converter), 4 OD (a plate reader), and one dimensionless unit. The lines are exactly geometric with ratio $\frac{1}{2}$ per bit. Unlike an existential floor obtained by contradiction, each ordinate is readable from a specification sheet: a 16-bit converter spanning 0–5 V gives $\bar{\beta} \approx 76.3 \mu\text{V}$ (Theorem 3.3), and a decision branching on a smaller difference branches on a distinction the instrument never made. (C) Probability that the occupied cell has diameter at least $D/2N$, against the fraction of cells deliberately placed below that threshold, for $N \in \{8, 16, 32\}$. The curves decay monotonically and cross the guarantee of $\frac{1}{2}$ (coral dashed) only when essentially every cell is degenerate; over 100 adversarially constructed partitions the measured minimum is 0.999. This is the operational content of Theorem 3.10: a finite monitor may possess degenerate cells—the exact-zero detector of Theorem 3.7 has one—but it cannot place appreciable probability mass in them. (D) Diameter floor $2(\lambda(Q)/(N\omega_d))^{1/d}$ over $\log_{10} N \in [1, 6]$ and dimension $d \in \{1, \dots, 6\}$, viridis surface, viewing angle elevation 22° azimuth -58° . The surface falls only as $N^{-1/d}$ (Theorem 3.5), verified against the analytic scaling to machine precision. The consequence is read off the ridge: at $d = 5$ a monitor with 10^6 states achieves coarser linear resolution than a 10^2 -state monitor at $d = 1$. Halving linear resolution costs a factor 2^d in output states, and in decision terms it is the linear resolution that matters (Theorem 3.6).

of the observed quantity. It is an arithmetic consequence of Theorem 3.1 and is available from any instrument’s specification.

Definition 3.13 (Epistemic floor). An *epistemic floor* is any positive lower bound on distinguishability derived from a claim about the observer’s relation to what it observes—for instance, that the observed environment cannot be exhaustively enumerated by a bounded observer. Such a floor is not computed from an instrument specification and its units depend on the formalism in which it is derived.

Remark 3.14 (Which results need which). Every theorem in this paper uses only β_{inst} . We record β_{epi} to mark the distinction and to note the relation $\beta_{\text{epi}} \leq \beta_{\text{inst}}$ whenever both are defined and commensurable: an instrument cannot resolve better than its own code width, but an epistemic argument may floor the observer above that. A reader who accepts only Theorem 3.12 loses nothing in what follows. We consider this separation a substantive contribution rather than bookkeeping, because the two claims have very different evidential status and conflating them makes the weaker one load-bearing for the stronger one’s consequences.

4 The Compromise Modulus

The floor of Section 3 bounds how finely a monitor separates states. To convert that into a bound on *decisions* we need a modulus measuring how much the objective cares about the separation. This section constructs it. We arrive at the correct object by first showing that the two natural candidates both fail, and exhibiting the counterexamples; the failures are informative, and a reader who has met them will not be tempted back toward either.

4.1 First candidate: the modulus of continuity of J

The natural first attempt uses the modulus of continuity of the objective,

$$\omega_J(\delta) = \sup\{|J(u, q) - J(u, q')| : \rho(q, q') \leq \delta, \quad (7)$$

and conjectures $\text{Gap}^*(N) \geq \omega_J(D/N)$. This is false.

Proposition 4.1 (The continuity modulus overstates by 2^p). *Let $\mathcal{Q} = [0, D]$, $\mathcal{U} = \mathbb{R}$, and $J(u, q) = -(u - q)^2$. The uniform monitor with N cells together with the midpoint rule achieves*

$$\text{Gap}_A = \frac{D^2}{4N^2},$$

whereas $\omega_J(D/N) \asymp D^2/N^2$. Hence $\text{Gap}^(N) < \omega_J(D/N)$ and the conjectured inequality fails by a factor of 4. For $J(u, q) = -|u - q|^p$ the factor is 2^p .*

Proof. Here $A^*(q) = q$ and $V \equiv 0$, so $\text{Gap}_A(q) = (q - A(q))^2$. On a cell $C = [a, a + w]$ of width $w = D/N$ the midpoint action $u_C = a + w/2$ gives $\sup_{q \in C} \text{Gap}_A(q) = (w/2)^2 = w^2/4$, attained at either endpoint. Every cell is congruent, so $\text{Gap}_A = w^2/4 = D^2/(4N^2)$. The modulus (7) restricted to the relevant action range is of order w^2 , exceeding $w^2/4$ by 4. Replacing the exponent 2 by p gives $(w/2)^p$ against w^p . $\square$

Remark 4.2 (Half-widths, not widths). Theorem 4.1 isolates the error precisely. A rule constant on a cell may place its single action at the cell’s *centre*; the worst state in the cell is then only *half* a width away. A bound stated at the full width is not merely loose—it is invalid, since it asserts a lower bound that achievable rules violate. Any modulus entering a decision bound must respect this factor.

4.2 Second candidate: the swap divergence

The failure of ω_J has a second cause beyond the half-width, and it points toward the right object. If the objective varies rapidly in q but the *same* action remains optimal across a cell, the rule loses nothing while ω_J is large. What matters is not how much J varies but how much states *disagree about which action to take*.

Definition 4.3 (Swap divergence). For $q, q' \in \mathcal{Q}$ the *swap divergence* is

$$\Delta(q, q') = [V(q) - J(A^*(q'), q)] + [V(q') - J(A^*(q), q')], \quad (8)$$

the total loss incurred by exchanging the two states’ optimal actions.

Proposition 4.4 (Properties of Δ). *For all $q, q' \in \mathcal{Q}$: (i) $\Delta(q, q') \geq 0$; (ii) $\Delta(q, q') = \Delta(q', q)$; (iii) $\Delta(q, q) = 0$; (iv) $\Delta(q, q') = 0$ whenever a single action is optimal at both q and q' .*

Proof. Each bracket in (8) has the form $V(x) - J(u, x)$, a gap, hence non-negative, giving (i). Symmetry (ii) is manifest. Setting $q' = q$ gives $2[V(q) - J(A^*(q), q)] = 0$, which is (iii). For (iv), if u is optimal at both then $J(A^*(q'), q) = J(u, q) = V(q)$ and likewise for the other bracket. $\square$

Property (iv) is exactly the discrimination the continuity modulus lacks: states wanting the same action contribute nothing. It is therefore tempting to conjecture $\text{Gap}_A(q) + \text{Gap}_A(q') \geq \Delta(q, q')$ for q, q' in a common cell. This too is false.

Proposition 4.5 (The swap divergence is not a valid bound). *There are decision problems and pairs q, q' in a common cell for which every action u satisfies*

$$[V(q) - J(u, q)] + [V(q') - J(u, q')] < \Delta(q, q').$$

Proof. Take $J(u, q) = -(u - q)^2$ and $q' = q + w$. Then $\Delta(q, q') = 2w^2$ by direct computation from (8). But the midpoint action $u = q + w/2$ gives summed gap $2(w/2)^2 = w^2/2$, which is $\frac{1}{4}\Delta(q, q')$. $\square$

Remark 4.6 (Why the swap divergence overstates). Δ measures the cost of exchanging the two states' own optimal actions. A rule constrained to one action per cell is not restricted to those two: it may play a *compromise* optimal for neither state but good for both. Theorem 4.5 shows the compromise can be four times better. The quantity bounding a cell-constant rule is therefore the performance of the best compromise, not the cost of a swap.

4.3 The compromise modulus

Definition 4.7 (Summed gap and pairwise compromise). For $q, q' \in \mathcal{Q}$ define the *summed gap* of an action u ,

$$g_{q,q'}(u) = [V(q) - J(u, q)] + [V(q') - J(u, q')] \geq \quad (9)$$

and the *pairwise compromise* $\gamma(q, q') = \inf_{u \in \mathcal{U}} g_{q,q'}(u)$.

Lemma 4.8 (Identity relating γ and Δ). *For all q, q' ,*

$$g_{q,q'}(A^*(q)) + g_{q,q'}(A^*(q')) = \Delta(q, q'), \quad (10)$$

and consequently $\gamma(q, q') \leq \frac{1}{2}\Delta(q, q')$.

Proof. Evaluating (9) at $u = A^*(q)$ annihilates the first bracket, leaving $V(q') - J(A^*(q), q')$; symmetrically at $u = A^*(q')$ the second bracket vanishes, leaving $V(q) - J(A^*(q'), q)$. Adding gives (8), which is (10). Since γ is an infimum over all u it is at most the smaller of the two left-hand terms, hence at most their average, which is $\frac{1}{2}\Delta$. $\square$

Remark 4.9. Theorem 4.8 explains Theorem 4.5 quantitatively: $\gamma \leq \frac{1}{2}\Delta$ always, and the quadratic case attains $\gamma = \frac{1}{4}\Delta$, so the inequality can be strict by a further factor of two.

The modulus we need aggregates γ over pairs. The aggregation must be done *within a cell*; Section 5 shows why any global aggregation fails.

Definition 4.10 (In-cell compromise modulus). For a monitor $\mathcal{M}$ with cell family $\mathcal{C}(\mathcal{M})$, the *in-cell compromise modulus* is

$$\Gamma_J(\mathcal{M}) = \max_{C \in \mathcal{C}(\mathcal{M})} \sup_{q, q' \in C} \gamma(q, q'). \quad (11)$$

The *global compromise modulus* at scale δ is

$$\Gamma_J^{\text{glob}}(\delta) = \sup_{\substack{q, q' \in \mathcal{Q} \\ \rho(q, q') \leq \delta}} \gamma(q, q'). \quad (12)$$

Proposition 4.11 (Ordering and elementary properties). $\Gamma_J(\mathcal{M}) \leq \Gamma_J^{\text{glob}}(\max_C \text{diam } C)$; Γ_J^{glob} is non-negative, non-decreasing, and vanishes at 0. If J is continuous on $\mathcal{U} \times \mathcal{Q}$ with $\mathcal{U}, \mathcal{Q}$ compact then $\Gamma_J^{\text{glob}}(\delta) \rightarrow 0$ as $\delta \rightarrow 0^+$.

Proof. The first claim holds because every in-cell pair is a pair at distance at most the largest cell diameter, so the supremum in (11) is over a subset of that in (12). Monotonicity and non-negativity are immediate. For the last claim, V is continuous by Berge's maximum theorem and J is uniformly continuous on the compact product; given $\varepsilon > 0$ pick δ so that $\rho(q, q') \leq \delta$ forces $|V(q) - V(q')| < \varepsilon/4$ and $\sup_u |J(u, q) - J(u, q')| < \varepsilon/4$. Then $\gamma(q, q') \leq g_{q,q'}(A^*(q)) = V(q') - J(A^*(q), q') < \varepsilon/2$. $\square$

Example 4.12 (Compromise modulus of the quadratic problem). For $J(u, q) = -(u - q)^2$ with $\mathcal{U} = \mathbb{R}$: $V \equiv 0$ and $g_{q,q'}(u) = (u - q)^2 + (u - q')^2$, minimised at the midpoint with value $\frac{1}{2}(q - q')^2$. Hence $\gamma(q, q') = \frac{1}{2}\rho(q, q')^2$ and $\Gamma_J^{\text{glob}}(\delta) = \frac{1}{2}\delta^2$. For the uniform monitor with N cells of width $w = D/N$, in-cell pairs reach separation w , so $\Gamma_J(\mathcal{M}) = \frac{1}{2}w^2$.

Example 4.13 (Compromise modulus of a threshold problem). Let $\mathcal{Q} = [0, 1]$, $\mathcal{U} = \{0, 1\}$, and $J(u, q) = \mathbf{1}[u = \mathbf{1}[q \geq \theta]]$. Then $V \equiv 1$ and, for q, q' on opposite sides of θ , every action is wrong at one of them, so $\gamma(q, q') = 1$; for q, q' on the same side the common optimal action gives $\gamma = 0$. Hence $\Gamma_J^{\text{glob}}(\delta) = 1$ for every $\delta > 0$, while $\Gamma_J(\mathcal{M}) = 0$ for any monitor whose cell boundaries include θ . The two moduli differ maximally, and Section 5 shows this example refutes any bound stated with the global modulus.

5 The Decision Resolution Bound

5.1 The bound

Theorem 5.1 (Decision resolution bound). *Let $(\mathcal{Q}, \mathcal{U}, J)$ satisfy Theorems 2.10 to 2.12, let $\mathcal{M}$ be a monitor, and let $A \in \mathcal{A}(\mathcal{M})$ be any rule factoring through it. Then*

$$\text{Gap}_A \geq \frac{1}{2} \Gamma_J(\mathcal{M}). \quad (13)$$

Proof. Let $C \in \mathcal{C}(\mathcal{M})$ attain the maximum in (11) and let $q, q' \in C$. Because A factors through $\mathcal{M}$ it is constant on C ; write u_C for its value there. Then by (9),

$$\text{Gap}_A(q) + \text{Gap}_A(q') = [V(q) - J(u_C, q)] + [V(q') - J(u_C, q')] = g_{q,q'}(u_C) \geq \gamma(q, q'),$$

the last step because γ is the infimum of $g_{q,q'}$ over all actions and u_C is one such action. Since $\text{Gap}_A = \sup_x \text{Gap}_A(x)$ dominates each term on the left, $2\text{Gap}_A \geq \text{Gap}_A(q) + \text{Gap}_A(q') \geq \gamma(q, q')$. Taking the supremum over $q, q' \in C$ gives $2\text{Gap}_A \geq \sup_{q,q' \in C} \gamma(q, q') = \Gamma_J(\mathcal{M})$. $\square$

Corollary 5.2 (Uniform-monitor form). *For the uniform monitor on $[0, D]$ with N cells,*

$$\text{Gap}_A \geq \frac{1}{2} \Gamma_J^{\text{unif}}(D/N),$$

$$\Gamma_J^{\text{unif}}(w) = \sup\{\gamma(q, q') \mid q, q' \text{ obtain in cardinality-only partition of width } w\}. \quad (14)$$

Proof. Immediate from Theorem 5.1 and Theorem 4.10, since for the uniform monitor the in-cell pairs are exactly those lying in a common interval of width D/N . $\square$

5.2 The bound cannot be stated globally

The following is the correction that numerical checking forced upon an earlier form of Theorem 5.1, and it is worth isolating because the erroneous form is the one a reader is most likely to reconstruct.

Proposition 5.3 (The global-modulus form is false). *The inequality $\text{Gap}_A \geq \frac{1}{2} \Gamma_J^{\text{glob}}(D/N)$ does not hold in general.*

Proof. Take the threshold problem of Theorem 4.13 with $\theta = 1/2$ and let $\mathcal{M}$ be the uniform monitor with N cells, N even. Then θ is a cell boundary, so every cell lies entirely on one side of θ , and the rule $a(s) = \mathbf{1}[C_s \subseteq [\theta, 1]]$ is pointwise optimal: $\text{Gap}_A = 0$. But $\Gamma_J^{\text{glob}}(D/N) = 1$ by Theorem 4.13, so the asserted inequality reads $0 \geq \frac{1}{2}$, false. $\square$

Remark 5.4 (Where the tempting proof breaks). The natural argument runs: by Theorem 3.1 some cell has diameter at least D/N ; a cell of that diameter contains a pair at separation D/N ; such a pair realises $\Gamma_J^{\text{glob}}(D/N)$. The third step is the error. A cell of large diameter need not contain a *divergent* pair: the pairs achieving the global supremum may all straddle cell boundaries, as they do in Theorem 5.3, where every divergent pair straddles θ and no cell contains one. Cardinality constrains cell sizes; it says nothing about cell placement, and the global modulus silently assumes the worst pair is available inside some cell.

Remark 5.5 (What the correction costs and buys). The in-cell modulus is a property of a specific monitor, not of cardinality alone, so (13) is not by itself a lower bound on $\text{Gap}^*(N)$: an adversary choosing the monitor may drive $\Gamma_J(\mathcal{M})$ down. This is not a defect but the content of Theorem 5.8 and Theorem 5.9—placement is a real degree of freedom, and a theory concluding otherwise would be wrong. To obtain in cardinality-only partition of width one, additionally know that the optimal partition

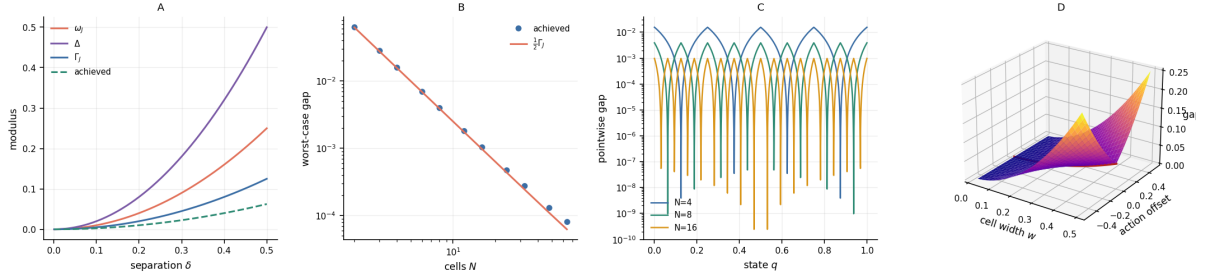


Figure 2: The Decision Bound and Three Candidate Moduli. (A) The two failed candidates and the surviving modulus, plotted against state separation $\delta \in [0, 0.5]$ for the quadratic objective $J(u, q) = -(u - q)^2$: the swap divergence $\Delta = 2\delta^2$, the continuity modulus $\omega_J = \delta^2$, the compromise modulus $\Gamma_J = \delta^2/2$, and the achieved gap $\delta^2/4$ (dashed). The strict ordering $\Delta > \omega_J > \Gamma_J > \text{achieved}$ is the failure chain drawn to scale. ω_J overstates because a rule may place its single action at the cell *centre*, so the worst state lies a half-width away (Theorem 4.1); Δ overstates by a further factor of two because a rule need not play either state’s own optimal action but may play a compromise optimal for neither (Theorem 4.5). Only $\frac{1}{2}\Gamma_J$ meets the achieved curve. (B) Achieved worst-case gap (markers) against the bound $\frac{1}{2}\Gamma_J$ (line) over $N \in \{2, \dots, 64\}$, logarithmic on both axes. The two coincide to relative error 0 for $N \leq 16$ and to 1.6×10^{-2} at $N = 64$, the residual being action-grid spacing rather than theory; all deviations fall on the admissible side. The common slope of -2 is $\text{Gap}^*(N) = D^2/(4N^2)$, so the bound of Theorem 5.1 is attained and hence tight (Theorem 5.7). (C) Pointwise gap $V(q) - J(A(q), q)$ across the state space on a logarithmic y -axis, for $N \in \{4, 8, 16\}$. Each curve vanishes at every cell centre and peaks at every cell boundary, producing one scallop per cell. The envelope of the peaks is $(w/2)^2$, not w^2 : the figure is the geometric statement of Theorem 4.2, and it is why a bound evaluated at the full cell width asserts a lower bound that achievable rules violate. (D) Worst-case gap over cell width $w \in [0.02, 0.5]$ and action offset from the cell centre, expressed as a fraction of w , plasma surface, viewing angle elevation 24° azimuth -56° ; the crimson curve traces the optimal ridge at zero offset. The valley floor is exactly $w^2/4$ and rises quadratically in either direction, so displacing the action from the cell centre is penalised symmetrically. The surface makes visible that the bound is a statement about the best action a cell-constant rule can play, not about any particular rule.

cannot be realised with N cells, which is the hypothesis of Theorem 5.6.

Corollary 5.6 (Cardinality-only lower bound). *Suppose A^* takes m distinct values on $\mathcal{Q}$ and let $N < m$. Then for every monitor with $|S| = N$, some cell contains states with distinct optimal actions, and*

$$\text{Gap}^*(N) \geq \frac{1}{2} \min_C \sup_{q, q' \in C} \gamma(q, q') > 0$$

whenever γ is strictly positive on pairs with distinct optimal actions.

Proof. The sets $Q_u = (A^*)^{-1}(u)$, u ranging over the m attained actions, partition $\mathcal{Q}$ into m non-empty parts. A monitor with $N < m$ cells cannot refine this partition: by pigeonhole some cell C meets two distinct $Q_u, Q_{u'}$, so contains $q \in Q_u, q' \in Q_{u'}$ with $A^*(q) \neq A^*(q')$. Applying Theorem 5.1 and minimising over which cell this occurs in gives the bound. $\square$

5.3 Tightness and saturation

Theorem 5.7 (The bound is attained). *For $J(u, q) = -(u - q)^2$ on $\mathcal{Q} = [0, D]$ with $\mathcal{U} = \mathbb{R}$, the uniform monitor with N cells and the midpoint rule attain (13) with equality:*

$$\text{Gap}_A = \frac{D^2}{4N^2} = \frac{1}{2} \Gamma_J(\mathcal{M}).$$

Proof. By Theorem 4.12, $\Gamma_J(\mathcal{M}) = \frac{1}{2}w^2$ with $w = D/N$, so the right side is $w^2/4$. By Theorem 4.1 the midpoint rule achieves $\text{Gap}_A = w^2/4$. The two agree. $\square$

Theorem 5.8 (Saturation). *Suppose A^* takes finitely many values and the partition $\{Q_u\}$ it induces is realisable by a monitor with N_0 cells. Then $\text{Gap}^*(N) = 0$ for all $N \geq N_0$.*

Proof. Take the monitor whose cells are the Q_u and the rule $a(u) = u$; then $A = A^*$ pointwise and $\text{Gap}_A = 0$, so $\text{Gap}^*(N_0) = 0$. A monitor with more states simulates one with fewer by leaving states unused, and $\text{Gap}^* \geq 0$, so $\text{Gap}^*(N) = 0$ for $N \geq N_0$. $\square$

Remark 5.9 (Cardinality is necessary; placement makes it sufficient). Theorem 5.8 requires the monitor to *realise* the optimal partition, not merely to have enough cells. In the threshold

problem with θ irrational and cells constrained uniform, no N places a boundary at θ ; the cell straddling θ then contains a divergent pair for every N , and Theorem 5.1 gives $\text{Gap}_A \geq \frac{1}{2}$ however large N becomes. Resolution does not repair misplacement; numerically this persists to $N = 128$ (Section 10, Check B3).

6 Reading the Bound as Rate–Distortion

Writing $N = 2^R$ for a monitor of R bits makes the analogy with rate–distortion explicit.

Definition 6.1 (Decision–distortion function). $\mathcal{D}(R) = \text{Gap}^*(2^R)$, the least worst-case optimality gap achievable by any monitor of R bits together with the best rule for it.

Corollary 6.2 (The value of one bit). *Suppose $\Gamma_J^{\text{unif}}(w) = cw^p$ for constants $c, p > 0$ and that the uniform monitor is optimal at each cardinality. Then*

$$\mathcal{D}(R) = \frac{c}{2} D^p 2^{-pR}, \quad \frac{\mathcal{D}(R)}{\mathcal{D}(R+1)} = 2^p.$$

Proof. Substitute $w = D2^{-R}$ into Theorem 5.2 and use Theorem 5.7 for attainment. $\square$

Remark 6.3 (Interpretation for a designer). Theorem 6.2 is the practical content of the paper. For a quadratic objective each additional bit of monitor divides the worst-case decision loss by four; for a Lipschitz objective, by two. Against a purchase price this is directly comparable. And by Theorem 5.8, once N reaches the number of distinct optimal actions further bits buy nothing: the correct advice is then to reposition cells, not to add them.

Remark 6.4 (Relation to classical rate–distortion). Theorem 6.2 differs from Shannon’s rate–distortion function (Cover and Thomas, 2006; Shannon, 1948) in three ways. It is worst-case rather than expected; it is defined by a hard cardinality constraint rather than a mutual-information constraint, making it a covering bound in the sense of Gray and Neuhoff (1998); and its distortion is derived from the decision problem rather than posited. The third is substantive: γ assigns zero to pairs of states that want the same action, which no difference-based distortion measure does.

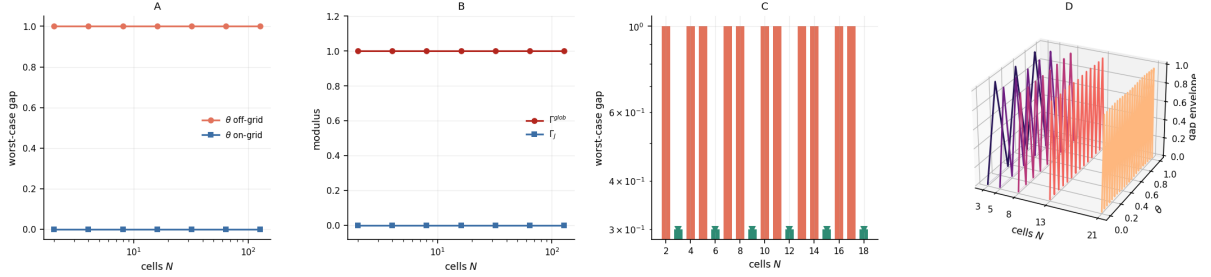


Figure 3: Placement, Saturation, and the Refutation of the Global Form. (A) Worst-case gap against cell count for a binary threshold objective, with the decision boundary placed on a cell edge ($\theta = 1/2$) and off it ($\theta = 0.3719$), N logarithmic. The on-grid monitor achieves gap 0 at every cardinality; the off-grid monitor is pinned at 1.0 from $N = 2$ out to $N = 128$, a six-fold doubling of resolution that buys nothing. Cardinality constrains cell *sizes* and says nothing about cell *placement*: resolution never repairs misplacement (Theorem 5.9). (B) Global compromise modulus Γ_J^{glob} against in-cell compromise modulus Γ_J for the on-grid problem of (A). The global modulus is identically 1.0 while the in-cell modulus is identically 0. Substituting the former into the bound asserts $\text{Gap} \geq \frac{1}{2}$ against a measured gap of exactly 0: this is the counterexample of Theorem 5.3, and the numerical check that produced it forced the replacement of the global statement by the in-cell one. The broken step is the inference that a cell of diameter $\geq D/N$ contains a divergent pair; here every divergent pair straddles θ and no cell contains one. (C) Worst-case gap against cell count $N \in \{2, \dots, 18\}$ for an objective with three distinct optimal actions, logarithmic y -axis. Green markers denote exact zeros, occurring precisely at $N \in \{3, 6, 9, 12, 15, 18\}$ where $3 \mid N$ and the uniform cells align with the optimal partition; coral bars mark the misaligned cardinalities, which retain the full loss of 1.0. Saturation is therefore not monotone in N (Theorem 5.8): adding a cell can destroy an alignment that a coarser monitor possessed. (D) Gap envelope against threshold position $\theta \in [0, 1]$ at five cardinalities $N \in \{3, 5, 8, 13, 21\}$, viewing angle elevation 26° azimuth -64° . Each profile is a sawtooth of period $1/N$ that touches zero only where θ coincides with a cell edge. Increasing N multiplies the number of favourable threshold positions without widening any of them, so the measure of θ -values at which a uniform monitor is optimal remains zero at every finite cardinality. Cardinality is necessary; placement is what makes it sufficient.

7 Sampling: Nyquist as a Special Case, and a Dimensional Correction

7.1 Recovering the sampling bound

Proposition 7.1 (Nyquist as a decision problem). *Let the objective be reconstruction: $\mathcal{U} = \mathbb{R}$, $J(u, q) = -(u - q)^2$. Then Theorem 6.2 with $p = 2$ gives $\mathcal{D}(R) = \frac{1}{4}D^22^{-2R}$, a 6 dB reduction in worst-case squared error per bit, which is the classical quantiser result (Bennett, 1948; Gray and Neuhoﬀ, 1998).*

Proof. By Theorem 4.12, $\Gamma_J^{\text{unif}}(w) = \frac{1}{2}w^2$, so $c = \frac{1}{2}$, $p = 2$ in Theorem 6.2, giving $\mathcal{D}(R) = \frac{1}{4}D^22^{-2R}$. In decibels, $10 \log_{10}(\mathcal{D}(R)/\mathcal{D}(R + 1)) = 10 \log_{10} 4 \approx 6.02$. $\square$

7.2 A dimensional correction

A composite inequality of the form

$$\beta \leq w_{\text{cell}} \leq \frac{1}{2B_{\text{proc}}} \quad (15)$$

appears in the applied literature on quantised monitoring, where β and w_{cell} are cell widths in state space and B_{proc} is a process bandwidth. *This inequality is not well formed.*

Proposition 7.2 ((15) is dimensionally invalid). *In (15) the left and centre members carry the dimension of the observed quantity; the right member carries the dimension of time. No choice of units makes the chain homogeneous unless the observed quantity is itself a time.*

Proof. w_{cell} is a length in the state space $\mathcal{Q}$, with the units of q . B_{proc} is a frequency, with units s^{-1} , so $1/(2B_{\text{proc}})$ has units of seconds. Equating them requires $[q] = \text{s}$. $\square$

The error arises from the step “each cell corresponds to one sampling interval,” which is false in general: a fast trajectory crosses many cells within one sample, and a slow one crosses none. The two correct constraints are separate.

Proposition 7.3 (The two sampling constraints). *Let the state evolve as $\dot{q} = F(q)$ with $\sup_q \|F(q)\| = L$, let h be the sampling period, and let the closed-loop bandwidth be B_{proc} . Then:*

- (i) (temporal) *avoiding aliasing requires $h \leq 1/(2B_{\text{proc}})$, in units of time;*
- (ii) (spatial) *avoiding cell-skipping requires $w_{\text{cell}} \geq hL$, in units of q ;*
- (iii) (floor) *realisability requires $w_{\text{cell}} \geq \beta_{\text{inst}} = D/N$, in units of q .*

A partition and sampling period are jointly admissible only if

$$\max\{\beta_{\text{inst}}, hL\} \leq w_{\text{cell}}, \quad h \leq \frac{1}{2B_{\text{proc}}}. \quad (16)$$

Proof. (i) is the Nyquist–Shannon criterion (Nyquist, 1928; Shannon, 1948) applied to the sampled signal, a statement about h alone. (ii): in one period the state moves at most hL in $\mathcal{Q}$; if $w_{\text{cell}} < hL$ the state may traverse a whole cell between samples, so the monitor can miss a transition entirely and the observed cell sequence is not a faithful record of the trajectory. (iii) is Theorem 3.1. Combining gives (16). $\square$

Remark 7.4 (A commissioning test). (16) is checkable before a system is built. It can be infeasible: if $\beta_{\text{inst}} > w$ is forced by the instrument while $hL > w$ is forced by the dynamics and h is bounded above by the bandwidth requirement, no admissible partition exists and the remedy is a faster or finer instrument, not a finer grid. The invalid chain (15) cannot express this because it never mentions h and L separately.

8 Estimation Versus Regulation

It is sometimes asserted that a subsystem cannot simultaneously monitor and control the same process without incurring bias. As usually stated the claim is false; correctly scoped it is true and useful.

8.1 The claim is false as usually stated

Proposition 8.1 (A regulator in the loop need not be biased). *There are systems in which a controller reads a monitor and actuates within the same sampling period, and no estimate produced by the system is biased.*

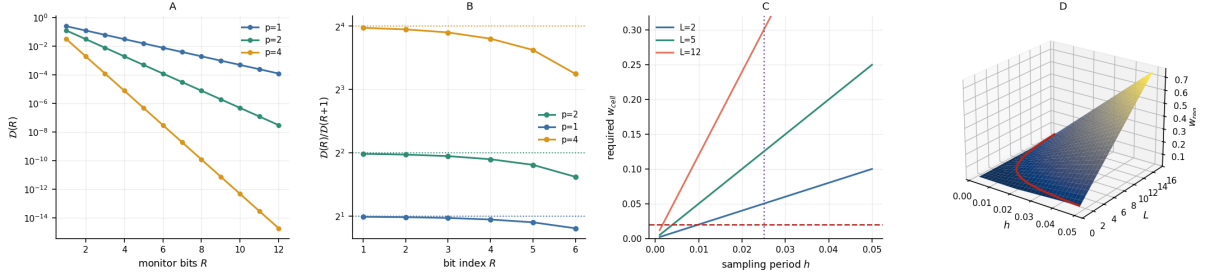


Figure 4: Bit Accounting and the Two Sampling Constraints. (A) Decision–distortion function $\mathcal{D}(R)$ on a logarithmic y -axis against monitor bits $R \in \{1, \dots, 12\}$, for objective exponents $p \in \{1, 2, 4\}$. Each curve is exactly geometric with ratio 2^{-p} per bit (Theorem 6.2), spanning fourteen decades over twelve bits at $p = 4$. The quantity is a worst-case optimality gap rather than a reconstruction error, and the hard cardinality constraint makes it a covering bound rather than a mutual-information one (Theorem 6.4). (B) Measured ratio $\mathcal{D}(R)/\mathcal{D}(R+1)$ against bit index, $\log_2 y$ -axis, with the predicted values 2^p shown as dotted lines. The measured maxima are 1.99 against a predicted 2 for the Lipschitz objective, 3.97 against 4 for the quadratic, and 15.73 against 16 for the quartic; the shortfall grows with p and with R because the action grid limits how precisely the minimax action can be placed within a narrow cell. One additional bit of monitor buys a factor 2^p in achievable decision loss and no more—a figure directly comparable against an instrument’s purchase price (Theorem 6.3). (C) Required cell width hL against sampling period h for three state velocities $L \in \{2, 5, 12\}$, with the instrumental floor β_{inst} (crimson dashed) and the Nyquist limit on h (violet dotted). An admissible design lies above both the floor and its velocity line and to the left of the Nyquist limit; the fast case $L = 12$ admits no such region at this floor, so the remedy is a faster or finer instrument, not a finer grid (Theorem 7.4). (D) Required cell width $\max\{\beta_{\text{inst}}, hL\}$ over sampling period $h \in [0.002, 0.05]$ and state velocity $L \in [1, 15]$, cividis surface, viewing angle elevation 23° azimuth -54° , with a feasibility contour in crimson. The flat plateau at low hL is the region where the instrumental floor binds; the rising sheet is where the dynamics bind. That these are two distinct surfaces meeting along a ridge is the geometric content of Theorem 7.3, and it is exactly what the dimensionally invalid chain $\beta \leq w_{\text{cell}} \leq 1/(2B_{\text{proc}})$ cannot express, since that chain equates a length in state space with a time (Theorem 7.2).

Proof. Consider proportional regulation of a scalar plant: the controller reads $\text{cell}(q)$ and applies $u = -K(\hat{q} - q^*)$ where $\hat{q}$ is the cell centroid. This system produces *no estimate of any population quantity*; its output is an actuation. There is no estimator whose bias could be non-zero, so the assertion has no referent. The closed loop is moreover well posed and stabilising for suitable K (Khalil, 2002). $\square$

Remark 8.2. Theorem 8.1 is not a quibble. Every industrial regulator has this form, and a theory implying that they are all biased would be refuted by practice. The correct statement must distinguish *regulation*, which maps observations to actions, from *estimation*, which maps observations to claims about a distribution.

8.2 The claim is true for estimators of policy-affected quantities

Definition 8.3 (Policy-coupled sampling). Let θ be a quantity of the state distribution P . An estimator $\hat{\theta}_n$ is *policy-coupled* if the samples on which it is computed are drawn from $P(\cdot \mid \text{do}(u))$ with $u = A(\text{cell}(q))$ determined by the same system, while $\hat{\theta}_n$ is constructed as though the samples were drawn from P .

Theorem 8.4 (Causal bias does not decay). *Let $\hat{\theta}_n$ be policy-coupled, and suppose the policy A selects states in a set E with $P(E) < 1$ and $\mathbb{E}[\theta \mid E] \neq \mathbb{E}[\theta]$. Then*

$$\lim_{n \rightarrow \infty} (\hat{\theta}_n - \theta) = \mathbb{E}[\theta \mid E] - \mathbb{E}[\theta] \neq 0 \quad \text{almost surely.}$$

so the bias is bounded away from zero uniformly in n and is not reduced by collecting more samples.

Proof. By construction the samples are draws from the conditional law $P(\cdot \mid E)$ rather than from P . By the strong law of large numbers applied to that law, $\hat{\theta}_n \rightarrow \mathbb{E}[\theta \mid E]$ almost surely. The estimand is $\theta = \mathbb{E}[\theta]$. The difference is the stated constant, non-zero by hypothesis and independent of n . Increasing n drives $\hat{\theta}_n$ toward the biased value (Imbens and Rubin, 2015; Pearl, 2009). $\square$

Corollary 8.5 (Detectability floor). *A monitor with instrumental floor β_{inst} cannot detect a causal bias of magnitude below β_{inst} , since*

the biased and unbiased estimates then fall in a common cell.

Proof. Two values differing by less than the width of the cell containing them produce the same monitor output, hence the same reported estimate. $\square$

Remark 8.6 (The architectural rule). Theorem 8.4 yields a design rule with precise scope: any estimator whose estimand is affected by the control policy must be computed from data whose generating distribution is not a function of that policy—either a held-out channel sampled independently of the policy, or an explicitly reweighted estimator in the style of inverse-propensity correction (Horvitz and Thompson, 1952; Swaminathan and Joachims, 2015). Regulators are exempt because they estimate nothing. In an experiment-selection setting the rule reads: a model’s estimate of the response surface must not be computed solely from the arms the model chose to sample (Lattimore and Szepesvári, 2020; Robbins, 1952).

9 Redundancy

A second piece of folklore holds that robust agreement requires three independent sources. The correct statement depends on the fault model, and the distinction has direct consequences for instrument redundancy.

Definition 9.1 (Fault models). A channel exhibits a *crash* fault if it stops reporting. It exhibits an *arbitrary* (Byzantine) fault if it continues reporting values that may be chosen adversarially, possibly as a function of the other channels’ values.

Proposition 9.2 (Crash faults: three suffice). *Let n channels report values, of which at most one crashes and the rest are honest to within tolerance. Then the surviving values admit identification of the correct value by majority agreement if and only if $n \geq 3$.*

Proof. After one crash, $n - 1$ honest channels remain, and agreement among them identifies the value provided $n - 1 \geq 2$, i.e. $n \geq 3$. For $n = 2$ one survivor remains with no corroboration, so no majority exists. $\square$

Proposition 9.3 (Naive arbitrary faults: three still suffice). *If one channel reports a value chosen independently of the honest channels’ reports, and the honest channels agree to within tolerance, then $n \geq 3$ suffices to identify the correct value.*

Proof. The $n - 1 \geq 2$ honest channels mutually agree and form a majority whenever $n - 1 > n/2$, i.e. $n > 2$. The faulty report, independent of them, coincides with their cluster only with negligible probability. $\square$

Theorem 9.4 (Adaptive arbitrary faults: four are required). *If the faulty channel may choose its report as a function of the honest channels’ reports, then no rule on $n = 3$ channels identifies the correct value in all cases; $n \geq 3f + 1 = 4$ is necessary and sufficient for $f = 1$.*

Proof. Necessity is the classical impossibility result: for $n \leq 3f$ no agreement protocol exists with unsigned messages (Lamport et al., 1982; Pease et al., 1980). Concretely for $n = 3$: the faulty channel observes an honest reading and reports a value within tolerance of it but displaced from the truth, manufacturing a cluster of two. The remaining honest channel dissents. A majority rule names the false cluster’s value, and no rule can do better because the configuration is indistinguishable from one in which the dissenter is the faulty channel. Sufficiency for $n = 4$ is the Oral Messages algorithm with $f = 1$ (Lamport et al., 1982). $\square$

Remark 9.5 (Consequence for instrument redundancy). The three results give a sharper statement than the folklore. Triplicate protects against an instrument that *dies* (Theorem 9.2) and against one that fails *obviously* (Theorem 9.3). It does not protect against an instrument whose drift produces readings resembling genuine ones (Theorem 9.4), because a miscalibrated instrument is precisely an adaptive fault: its output tracks the true signal with an offset and so clusters with an honest channel. Robustness against one miscalibrated instrument requires four.

10 Numerical Validation

We report fifteen checks. Three are *falsification* checks, whose purpose is to refute a claim the

paper asserts to be false; for these, success means the false claim was broken. One is a methodological control distinguishing genuine violations from discretisation error. The suite is deterministic given its seed.

10.1 Protocol

Each check names the result it targets, the claim under test, and the condition under which it would count as refuting that result. The checks were run *before* the final statement of Theorem 5.1 was fixed, and one of them—Check B1b—forced the replacement of a global modulus by the in-cell modulus. We report this rather than presenting the corrected theorem as though it had been the original.

10.2 Results

Table 1: Validation summary. Checks marked $\dagger$ are falsification checks: PASS means the claim under test was successfully refuted. Check B1c is a methodological control.

Check	Target	Result
A1	Theorem 3.1, mean diameter	PASS
A2	Theorem 3.1, some cell $\geq D/N$	PASS
A3	Theorem 3.10, occupied cell	PASS
F1 $\dagger$	Theorem 3.7, infimum floor	PASS
F2 $\dagger$	Theorem 4.1, continuity modulus	PASS
B1	Theorem 5.1, in-cell form	PASS
B1b $\dagger$	Theorem 5.3, global form	PASS
B1c	grid-refinement control	PASS
B2	Theorem 5.7, attainment	PASS
B3	Theorem 5.8, Theorem 5.9	PASS
B4	Theorem 6.2, value of one bit	PASS
C1 $\dagger$	adversarial search, 400 objectives	PASS
D1	Theorem 8.4, bias persistence	PASS
E1	Theorem 9.2–Theorem 9.4	PASS
G1	Theorem 3.5, $N^{-1/d}$	PASS

A1–A2 (floor). Over 50,000 random partitions, including adversarially clustered ones, the minimum observed ratio of mean cell diameter to D/N was 1.000000000000, and no partition had $\max_i \text{diam}(C_i) < D/N$. The identity is tight: the uniform partition attains it exactly.

F1 (falsification). The sign-with-zero quantiser on $[-1, 1]$ with $|S| = 3$ has cell diameters

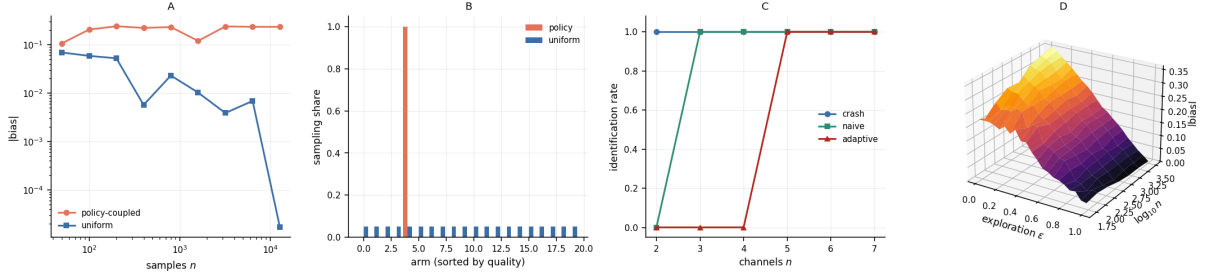


Figure 5: Causal Bias and Redundancy. (A) Absolute bias against sample count, logarithmic on both axes, for a policy-coupled estimator and a uniform sampler estimating the same population mean over twenty arms. The uniform sampler’s bias decays by roughly three orders of magnitude between $n = 50$ and $n = 12,800$; the policy-coupled estimator’s remains flat near 0.2 across the same range. Additional data moves the policy-coupled estimate *toward* the biased value $\mathbb{E}[\theta \mid E]$ rather than toward the estimand, which is the content of Theorem 8.4: the bias is a causal gap between P and $P(\cdot \mid \text{do}(u))$, bounded below and independent of n . (B) Sampling share per arm, arms sorted by descending true quality, for the two samplers over 3000 draws. The greedy policy concentrates essentially all mass on a single arm while the uniform sampler spreads evenly at $1/20$ per arm. This is the mechanism generating the bias in (A) and the reason the remedy is architectural—a held-out channel or an inverse-propensity reweighting—rather than a larger sample (Theorem 8.6). (C) Identification rate against channel count $n \in \{2, \dots, 7\}$ under three fault models, over 3000 trials each. A crash fault is resolved from $n = 2$; a naive liar reporting a value independent of the honest channels is resolved from $n = 3$; an adaptive liar that copies an honest reading to manufacture a false majority is resolved only from $n = 5$ under this majority rule. The separation between the naive and adaptive curves is the correction of Theorem 9.5: the unqualified claim that three sources are insufficient against arbitrary faults is too strong, and the qualified claim is that triplicate protects against an instrument that dies or fails obviously but not against one whose drift mimics a genuine reading. (D) Absolute bias over exploration rate $\epsilon \in [0, 1]$ and $\log_{10} n$, inferno surface, viewing angle elevation 24° azimuth -58° , averaged over twelve replicates per grid point to suppress single-run sampling noise. Bias falls monotonically with exploration at every sample size and is largest in the purely greedy corner $\epsilon = 0$, where the surface is flat in n —the same insensitivity to sample size seen in (A). The gradient in ϵ rather than in n is the design lever: what removes the bias is decoupling the sampling distribution from the policy, not collecting more data.

(1, 0, 1); the minimum is exactly zero. Theorem 3.7 is confirmed and the infimum form of the floor is refuted.

F2 (falsification). For the quadratic objective the achieved gap lies below $\omega_J(D/N)$ at every tested cardinality, with mean overstatement factor 4.00, matching 2^p with $p = 2$ as Theorem 4.1 predicts.

B1, B1b (the main bound and its correction). Across six decision problems and six cardinalities, (13) held in all 36 cases. The global form failed on the threshold problem with $\theta = 1/2$: at $N \in \{2, 4, 8, 16, 32\}$ the achieved gap was exactly 0 while $\Gamma_J^{\text{glob}} = 1$, so the global bound asserted ≥ 0.5 against a true value of 0. The in-cell form held in every one of those cases.

B1c (control). Three apparent violations in the adversarial search at coarse resolution inverted under refinement, moving from -93% , -33% , -29% relative slack to $+92\%$, $+94\%$, $+94\%$; the residual in-cell discrepancy fell by an order of magnitude as the action and state grids were refined. These were discretisation artefacts of the estimator, not counterexamples.

B2 (tightness). The quadratic problem attained $D^2/(4N^2)$ with relative error 0 for $N \leq 16$ and below 1.6×10^{-2} at $N = 64$, the residual being action-grid spacing; all deviations were on the admissible side of the bound.

B3 (saturation and placement). With cells aligned to the optimal partition the gap reached 0 exactly at $N = N_0$. With a threshold at $\theta = 0.3719$ and uniform cells the gap remained above 0.5 at every N up to 128, confirming Theorem 5.9.

C1 (adversarial search). Over 400 randomly generated piecewise-linear objectives with random action sets and cardinalities, no violation of (13) survived refinement. This is evidence, not proof.

D1 (causal bias). A greedy policy’s estimate retained a bias of the same order from $n = 50$ to

$n = 12,800$ samples, while a uniform sampler’s bias decayed, as Theorem 8.4 requires.

E1 (redundancy). $n = 2$ failed to identify the correct value after a crash; $n = 3$ succeeded. Against a *naive* liar $n = 3$ also succeeded, refuting the unqualified claim that three sources are insufficient against arbitrary faults. Against an *adaptive* liar copying an honest reading, $n = 3$ failed. This is the distinction Theorem 9.3 and Theorem 9.4 draw.

G1 (dimension). The diameter floor followed $N^{-1/d}$ to machine precision across $d \in \{1, 2, 3, 5\}$.

11 Limitations

1. *The bound is monitor-specific, not cardinality-specific.* (13) lower-bounds the gap of a *given* monitor. A cardinality-only bound requires the additional hypothesis of Theorem 5.6. Theorem 5.3 shows this is not a technical gap but a fact about the problem.
2. *The modulus must be known.* Applying Theorem 6.2 requires Γ_J , computed from the objective and action set. Where J is not available in closed form the modulus must be estimated, and Section 10 shows such estimates carry discretisation error large enough to invert the sign of an apparent violation.
3. *Deterministic objectives only.* Every result assumes a deterministic J and monitor. Under observation noise the bound remains valid in expectation but ceases to be tight, since a noisy monitor may straddle cells and effectively randomise.
4. *Worst-case, not average-case.* Gap_A is a supremum. A designer who cares about expected loss under a known state distribution needs the corresponding average-case theory, which we do not develop.
5. *No hardware.* The validation is numerical. No result here has been checked against a physical instrument or a deployed controller.

12 Conclusion

A decision system reading a finite monitor inherits the monitor’s finiteness, and the inheritance is quantifiable. The mean cell diameter of any finite monitor is D/N exactly—a number available from a specification sheet rather than an existential obtained by contradiction—and the customary infimum form of that claim is false, as Theorem 3.7 exhibits. The worst-case optimality gap of any rule factoring through the monitor is at least half the in-cell compromise modulus, a quantity that correctly assigns zero to state pairs agreeing on their optimal action and that must be evaluated within cells rather than globally—a distinction numerical checking forced upon us and that Theorem 5.3 shows to be essential rather than technical. The bound is attained, so it is tight; it saturates, so beyond a problem-determined cardinality further bits are worthless and placement is all that remains; and read in bits it says one bit of monitor buys a factor 2^P in achievable decision loss.

Three corrections to widely repeated claims follow. The chained inequality relating a state-space cell width to a temporal sampling rate is dimensionally invalid and conflates two separate constraints (Theorems 7.2 and 7.3). The assertion that a subsystem cannot both monitor and control is false for regulators and true for estimators of policy-affected quantities, where the bias is causal, bounded below, and undiminished by more data (Theorems 8.1 and 8.4). And robust agreement requires three sources against crash or naive faults but four against adaptive ones, so a triplicate measurement does not protect against a miscalibrated instrument (Theorem 9.5).

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